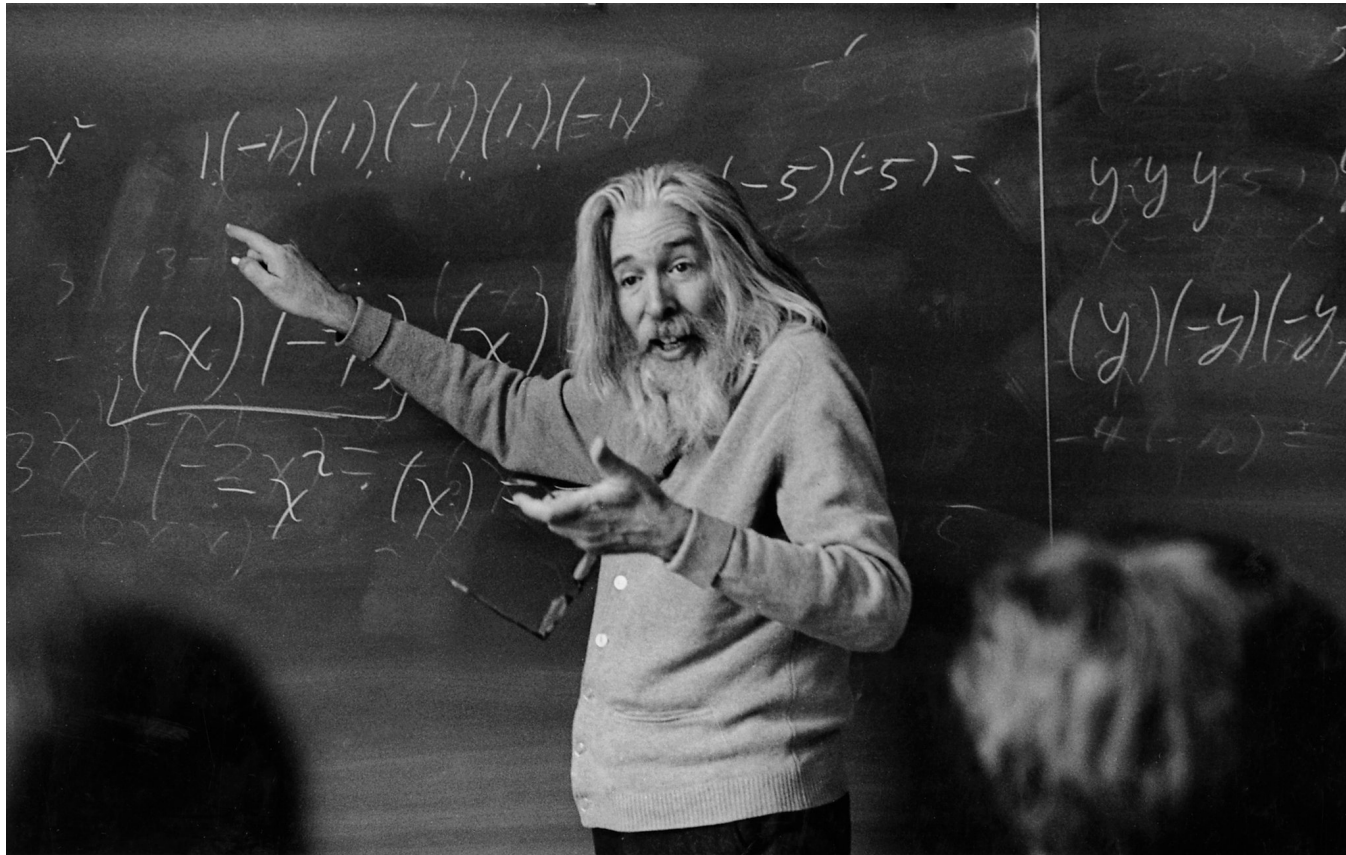


Raymond Smullyan's Tableaux First-Order Logic



Courtesy of Luciano Serafini (FBK, Trento)

Reasoning tasks in FOL

In First order logics we have the same reasoning task as in propositional logics (and any other logics)

Model checking

For a closed formula ϕ check if $\mathcal{I} \models \phi$

Satisfiability

Find an interpretation $\mathcal{I}$ that satisfies a closed formula ϕ . I.e., check if there is a $\mathcal{I}$ such that $\mathcal{I} \models \phi$.

Validity

Check if a formula ϕ is valid, i.e., if for all interpretations $\mathcal{I}$, $\mathcal{I} \models \phi$

Logical consequence

Check if a formula ϕ is a logical consequence of a set of formulas Γ , i.e., $\Gamma \models \phi$

Tableaux Calculus

- The Tableaux Calculus is an algorithm solving the problem of satisfiability.
- If a formula is satisfiable, then there exists an open branch in the tableaux of this formula.
- the procedure attempts to construct the tableaux for a formula. Sometimes it's not possible since the model of the formula is infinite.
- The basic idea is to incrementally build the model by looking at the formula, by decomposing it in a top/down fashion. The procedure exhaustively looks at all the possibilities, so that it can possibly prove that no model could be found for unsatisfiable formulas.

Semantic tableaux

Definition

A tableau is a rooted tree, where each node carries a first order sentence (closed formula), and the children of a node n are generated by applying a set of **expansion rules** to n or to one of the ancestors of n .

Definition

The expansion rules for a first order semantic tableaux are those for the propositional semantic tableaux, extended with the following rules that deal with the quantifiers:

$$\gamma \text{ rules} \quad \frac{\forall x.\phi(x)}{\phi(t)} \quad \frac{\neg\exists x.\phi(x)}{\neg\phi(t)} \quad \text{Where } t \text{ is a term free for } x \text{ in } \phi$$

$$\delta \text{ rules} \quad \frac{\neg\forall x.\phi(x)}{\neg\phi(c)} \quad \frac{\exists x.\phi(x)}{\phi(c)} \quad \text{where } c \text{ is a new constant not previously appearing in the tableaux}$$

Tableaux production rules for propositional logic

... for propositional connectives

α rules	$\frac{\phi \wedge \psi}{\phi}$ ψ	$\frac{\neg(\phi \vee \psi)}{\neg\phi}$ $\neg\psi$	$\frac{\neg\neg\phi}{\phi}$	$\frac{\neg(\phi \supset \psi)}{\phi}$ $\neg\psi$	
β rules	$\frac{\phi \vee \psi}{\phi \mid \psi}$	$\frac{\phi \supset \psi}{\neg\phi \mid \psi}$	$\frac{\neg(\phi \wedge \psi)}{\neg\phi \mid \neg\psi}$	$\frac{\phi \equiv \psi}{\phi \mid \neg\phi}$ $\psi \mid \neg\psi$	$\frac{\neg(\phi \equiv \psi)}{\phi \mid \neg\phi}$ $\neg\psi \mid \psi$

Substitution $\phi[x/t]$

If $\phi(x)$ is a free variable and t is a term, we use the notation $\phi(t)$ instead of the more precise notation $\phi[x/t]$ to represent the substitution of x for t in ϕ .

Substitution

$\phi[x/t]$ denotes the formula we get by replacing each free occurrence of the variable x in the formula ϕ by the term t . This is admitted if t does not contain any variable y such that x occurs in the scope of a quantifier for y (i.e., in the scope of $\forall y$ or $\exists y$).

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is not free for y in $\forall x. P(x, y)$

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is not free for y in $\forall x. P(x, y)$

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is not free for y in $\forall x. P(x, y)$

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is not free for y in $\forall x. P(x, y)$

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is not free for y in $\forall x. P(x, y)$

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is not free for y in $\forall x. P(x, y)$

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is
not free for y in $\forall x. P(x, y)$

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is not free for y in $\forall x. P(x, y)$

Substitution $\phi[x/t]$

Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since $f(x)$ is
not free for y in $\forall x. P(x, y)$